

Potential MSc Adaptive Cybersecurity supervisors

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Prof Michael Madden

Thematic Area Keywords:

Machine Learning, Data Analysis, Probabilistic Reasoning, Reinforcement Learning, Deep Learning, Foundation Models.

Possible Problem Domains:

Applications in computer vision, robotics, autonomous vehicles, engineering, physical sciences and medicine.

Project ideas involving machine learning, including deep learning and classical ML approaches, in application domains including engineering, medicine and science, are always welcome.

Links:

Google Scholar: https://scholar.google.com/citations?user=M2_AkRIAAAAJ

University of Galway: <https://research.universityofgalway.ie/en/persons/michael-madden/>

Examples of Previous Projects Supervised:

- *Large Language Models for Extracting Structured Information from Unstructured Medical Notes (with University Hospital Galway)*
- *Analysing Biomarkers of Heart Disease Using Deep Learning for Tabular Data (with University Hospital Galway)*
- *Deep Learning to Estimate a Person's Measurements from a Photograph (with a Galway startup company)*
- *Effect of Adverse Weather Conditions on Computer Vision for Automated Driving (with Valeo Vision Systems)*
- *Fooling Deep Neural Networks for Traffic Sign Recognition*
- *Using Virtual Reality Environments to Generate Automatically Annotated Training Images for Deep Learning*
- *Identification of Universal Challenges in Visual Perception for Autonomous Vehicles (with Valeo Vision Systems)*
- *Using Deep Learning to Diagnose and Classify Skin Cancer*
- *Parking Spot Detection for Autonomous Driving using Computer Vision and Machine Learning*
- *Combining Synthetic and Real Data in Deep Learning for Situations where Real Data is Hard to Collect*
- *Semantic Annotation of Natural Images Using Deep Learning and Synthetic Training*
- *Hardware and Deep Learning Solution for Tallying Votes in Irish Elections*

Robot with AI and Computer Vision to Solve Jigsaws

Dr Peter Paul Buitelaar

Thematic Area Keywords:

Natural Language Processing

Possible Problem Domains:

knowledge graph extraction from text and their use in NLP, e.g. term/relation extraction, multimodal knowledge graphs, use of knowledge graphs in chatbot or other NLP development

Links: (e.g., Google scholar profile or homepage)

<https://scholar.google.com/citations?hl=en&user=aDJBCh4AAAAJ>

Examples of Previous Projects Supervised:

- Predicting Company Success and Employee Satisfaction Using Glassdoor Reviews: An NLP and Knowledge Graph Approach
- Explainable News Recommendation Using LLMs and Knowledge Graphs
 1. Enhancing Aspect-Based Sentiment Analysis with ConceptNet

Dr Enda Barrett

Thematic Area Keywords:

Research interests include:

Applications of artificial intelligence and machine learning Cybersecurity Cloud Computing Energy Management HVAC Optimisation Reinforcement Learning Distributed Computing

For details visit:

<https://research.universityofgalway.ie/en/persons/enda-barrett/>

or

Google Scholar:

<https://scholar.google.com/citations?user=5rSy7vgAAAAJ&hl=en&oi=ao>

Possible Problem Domains:

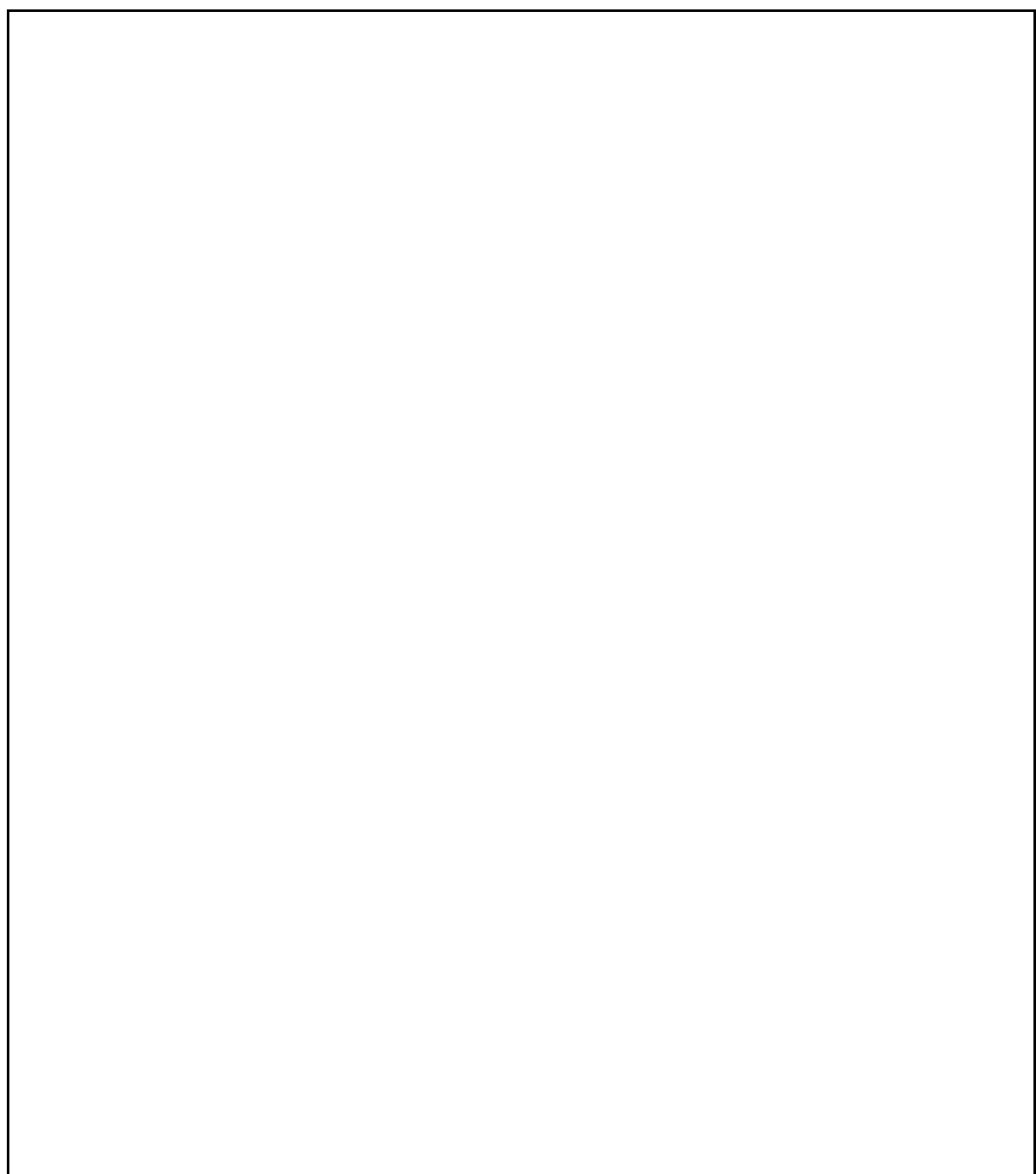
Applications of artificial intelligence and machine learning for:

- Cybersecurity
- Cloud Computing
- Energy Management HVAC Optimisation
- Reinforcement Learning
- Distributed Computing

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Examples of Supervised MSc Projects:

1.



Dr. Malika Bendeche

Thematic Area Keywords:

Artificial intelligence (AI), Machine Learning, Big Data Analytics, Data Governance (data quality, data value, data protection/privacy, data ethics), AI Governance, Ethical & Trustworthy AI.

Possible Problem Domains:

- Application of machine learning and data analytics in Cybersecurity
- Privacy and trustworthiness of Analytics/Machine Learning techniques.
- Bias in AI systems.
- Data governance and AI governance.

Links: You can find some of my recent publications via my Google scholar profile.

<https://scholar.google.com/citations?user=032sCkEAAAAJ&hl=en>

Examples of Previous Projects Supervised:

- Building Trustworthy Intrusion Detection Systems: Explainable and Robust Deep Learning Approaches.
- Ensuring Fairness and Transparency in Cyber Risk Scoring Models
- Ethical and trustworthy AI in the public sector.
- Designing a Privacy-Preserving Machine Learning Framework Using Federated Learning.

Dr. Joseph Lemley

Thematic Area Keywords:

Artificial Intelligence, Computational Intelligence, Image Processing, Machine Learning, Computer Vision

Possible Problem Domains:

Exploring the potential of novel sensors in AI applications.
Fostering research collaborations with universities and researchers globally.
Contributing to advancements in vehicular safety, sensing technology, and AI.

Keywords: Deep Learning, Driver Monitoring, Sensor Fusion, Computer Vision, Artificial Intelligence, Neural Networks, Machine Learning, Research & Development, Government Funding, University Collaboration, Level 3 and 4 autonomous driving

Links:

<https://scholar.google.co.in/citations?hl=en&user=ciZE8sIAAAAJ>

Dr. John McCrae

Thematic Area Keywords:

Natural Language Processing, Knowledge Graphs, Linked Data, Machine Learning, Under-resourced Languages, Lexicography

Possible Problem Domains:

Adapting Large Language Models to (Hindi/Tamil/Yoruba/etc.)

Large Language Models for Non-Standard Dialects
 Dynamic Recursive Transformer
 Improving Synthetic Data Quality for Text Classification
 Predicting Morphological Relations from Embeddings
 Rewording Gender-Biased Text with LLMs
 When not to link? - Finding Unmatchable Entities in Knowledge Graphs

Links:

<https://john.mccr.ae/>

Examples of Previous Projects Supervised:

- *A survey of link prediction combining explicit and latent features over heterogeneous networks*
- *A comparative study of SVN and LSTM deep learning algorithms for stock market prediction*
- *Neologism Detection in Social Media*
- *Teanga: A Linked Data based platform for Natural Language Processing*
- *Semantic Tagging of Complex Tabular Renderings of Financial PDF Documents*
- *Nineteenth-Century Probabilistic Steamship Data Disambiguation*

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Dr. Jawad Manzoor

Thematic Area Keywords:

Computer Network Protocols, Network Security, Blockchain Security, IoT

Possible Problem Domains:

- Security evaluation of private blockchains
- Evaluation of security features of modern application and transport layer protocols
- Collaborative anomaly detection using Federated Learning
- Improving protection against adversarial attacks using Machine Learning

Links:

https://scholar.google.com/citations?hl=en&user=1ZCNYysAAAAJ&view_op=list_works&sortby=pubdate

(Remark: Dr Manzoor joined the School of CS at University of Galway in early January. The link above may still show a previous affiliation.)

Examples of Previous Projects Supervised:

Examples of Previous Projects Supervised:

- Evaluation of open-source SIEM solutions
- Augmenting SIEM systems with threat intelligence
- Anonymous Tor traffic detection
- Network traffic classification using Machine Learning
- Insider threat detection

Supervisor Name: Dr. Matthias Nickles

Thematic Area Keywords:

Agents and Multi-Agent Systems.
Knowledge Representation and Reasoning (Formal Logic, Satisfiability Solving, Answer Set Programming, Knowledge Graphs).
Machine Learning (Reinforcement Learning, Federated Learning).
Small and Large Language Models.

Possible Problem Domains:

Human-AI and Multi-Agent interaction. Knowledge Bases. Agentic AI for, e.g., Software Development. Language Model Tuning and Memory. AI for CyberSecurity.

Links with further information about my research:

<https://research.universityofgalway.ie/en/persons/matthias-nickles/>

Examples of Previous Projects Supervised:

- *Opinion Retrieval: Leveraging Instruction Fine-Tuning and Supervised Contrastive Learning in Large Language Models*
- *Search Space Visualisation for Satisfiability Solvers*
- *Adaptive Autoencoder-Based Federated Learning for Malware Detection*
- *Evaluating the Electricity Efficiency of LoRA-based Fine-Tuning Techniques in Binary Document-Level Sentiment Classification*
- *A Hybrid LLM and GNN Approach to Banking Fraud Detection*

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Thematic Area Keywords:

Software Engineering and Testing, Cloud Computing, Smart Cities, Economics, Learning Enhancement, Artificial Intelligence, Optimisation

Possible Problem Domains:

Software Engineering: Mining Software Repositories (e.g., World of Code and CodeNet).
Generative Software Engineering/Testing.

Complex Systems: Auto-ML, Model-Driven Development/Testing of ML Systems

Cloud Computing: Optimising Applications/Communications in Low Earth Orbit Satellites

Smart Cities: Learning Efficient Traffic Management Systems

Economic: Explainable AI for Economic Growth Data, Identify Drivers for Regional Economic Integration

Education: Improve Engagement and Inclusion in Teaching Slide with Generative AI

Links: (e.g., Google scholar profile)

- <https://scholar.google.com/citations?user=aevYq0oAAAAJ>
- <https://takfarinassaber.github.io/>

Dr. Michael Schukat

Thematic Area Keywords:

Artificial Intelligence, Embedded Systems, Computer Vision, Distributed Computing, Cybersecurity, Health Informatics

Possible Problem Domains:

Application of machine learning, particularly reinforced learning and deep neural networks, to areas including cybersecurity (e.g., intrusion detection), computer vision (e.g., feature recognition, scene understanding and synthetic data generation), embedded systems (e.g. DNN implementation on resource-constrained devices), and healthcare (e.g., biosignal analysis)

Links: You can find some of my recent publications via my Google scholar profile:

https://scholar.google.com/citations?user=z_3p87AAAAAJ&hl=en&oi=ao

Examples of Previous Projects Supervised:

- *A Docker-based Solution for Deep Learning Researchers (company collaboration)*
- *Toolkit for Extracting Data from Videos based on Face Recognition in Videos (company collaboration)*
- *Learnable Augmentation - Expanding on Smart Augmentation Techniques (company collaboration)*
- *Synthetic Face-ID Generation using Generative Adversarial Networks (company collaboration)*
- *Adversarial Audio Synthesis for Child Data (company collaboration)*
- *New Framework for Adaptive and Agile Honeypots*
- *Machine Learning in Power Management*
- *Application of the k-nearest neighbours algorithm for Intrusion Detection in Industrial Control Systems*
- *Sensor Fusion of Thermal and Event Cameras with Health Sensors for Health Metrics Estimation (company collaboration)*
- *Utilising Transfer Learning and Deep Q Learning for HVAC Control*

Dr. Finlay Smith

Thematic Area Keywords:

Genetic Algorithms, Adaptive Systems, Fuzzy Logic, Very Large Numbers

Possible Problem Domains:

Adaptive game playing, optimising factors for Toom-Cook multiplication

Examples of Previous Projects Supervised:

- Priority based solver for a real-time dynamic vehicle routing
- Adapting in an Uncertain World"
- Optimisation enhancement using self-organising fuzzy control
- [Combining symbolic conflict recognition with markov chains for fault identification](#)
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Dr. Ihsan Ullah

Thematic Area Keywords:

Computer Vision, Deep Learning, Explainable-AI, Federated Learning, Differential Privacy

Google Scholar (for detail and recent results look at recent paper published):

https://scholar.google.co.in/citations?hl=en&pli=1&user=v6N_RLkAAAAJ

Possible Problem Domains:

I am interested in using deep learning models for:

- Deep Fake Recognition
- Adaptive Intrusion Detection and Deception System

In addition, I am also interested in:

- Federated learning in health domain
- Differential privacy for machine learning models
- Adversarial Camouflage in Autonomous Vehicles
- Understanding deep learning models with explainable-AI techniques for explaining models designed for health diagnosis from sensors data

Or if you want to discuss a potential idea for your thesis that is similar to my research interests, we can discuss that as well. Email me at ihsan.ullah@universityofgalway.ie

Examples of Previous Projects Supervised/Involved:

- Pavement surface classification, Segmentation, and rating (<https://www.paveanalytics.eu/>)
- Robust Monocular Depth Estimation Against Adversarial Camouflage in Autonomous Vehicles (with Valeo Vision Group)
- Privacy-Preserving Video Human Activity Recognition
- Synthetic data generation for Cardiovascular CT Images
- Emotion recognition with deep learning (using images, videos, &/or health sensor data)
- [Enhancing semantic segmentation of aerial images with inhibitory neurons in deep learning models](#)
- [Explaining deep learning models for structured data and detecting anomalies in image datasets](#) (using tabular &/or image data)
- Graph Convolutional neural networks: analysis, improvements, and results (using graphs)
- Person Re-Identification using deep learning model (using images)
- Epilepsy detection with deep learning (using EEG signals)
- Action recognition using deep learning models (using videos)
- Object recognition using deep learning models (using images)
- Identifying the Level of Diabetic Retinopathy Using deep learning models (using images)
- Body condition scoring of cows using deep learning (using images &/or videos)

Dr. Priyanka Verma

Thematic Area Keywords:

AI/ML, AI and Sustainability, Onehealth (Human, Animal and Agriculture), Cyber Security, Industry 5.0, Federated Learning

Possible Problem Domain:

Project ideas focusing on developing sustainable AI and next-generation AI-driven solutions, with applications in cybersecurity, healthcare, smart technologies.

Links: (e.g., Google scholar profile)

Google scholar : <https://scholar.google.co.in/citations?user=BNKQ-UgAAAAJ&hl=en>

University of Galway: <https://research.universityofgalway.ie/en/persons/priyanka-verma/>

Personal website: <https://priyankaverma.info/>

Examples of Previous Projects Supervised:

- Applications of AI in Health Activity Recognition (HAR)
- A Secure Adversarial Attack Detector Framework for Monitoring Network Intrusion in IoT Environment
- Predicting Cybersecurity Threats: A Machine Learning Approach to Log Analysis and Risk Scoring
- AI-Driven Cyber Resilience: Leveraging Machine Learning and Deep Learning for Proactive Threat Detection and Risk Mitigation
- AutoML for Cardiovascular Disease Prediction: Bridging AI and Clinical Practice
- Towards Explainable and Accessible AI Frameworks for Alzheimer's Disease Diagnosis
- GCGConv: Enhancing Human Activity Recognition with Global Context Graph Convolutions
- Leveraging transfer learning domain adaptation model with federated learning to revolutionise healthcare

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Google scholar : <https://scholar.google.co.in/citations?user=BNKQ-UgAAAAJ&hl=en>

University of Galway: <https://research.universityofgalway.ie/en/persons/priyanka-verma/>

Personal website: <https://priyankaverma.info/>

Examples of Previous Projects Supervised:

- Applications of AI in Health Activity Recognition (HAR)
- A Secure Adversarial Attack Detector Framework for Monitoring Network Intrusion in IoT Environment
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- GCGConv: Enhancing Human Activity Recognition with Global Context Graph Convolutions
- Leveraging transfer learning domain adaptation model with federated learning to revolutionise healthcare

Dr. Waqar Shahid Qureshi

Thematic Area Keywords:

Mobile Robotics, Smart Agriculture Technologies, Computer Vision, Image Processing for Intelligent Systems, Cyber-Physical System, Deep learning models for image analysis, Reinforcement Learning

Possible Problem Domain: Object Détection, Image Pixel Segmentation, Object Tracking, Explainable Deep learning models, Application of Machine Learning or Deep learning in Aerial Imagery of crops, fruit counting, or natural enemies detection, Application of image analysis for pavement surface analysis, pavement surface type, and pavement layer detection using classical machine learning or deep learning techniques.

Links: (e.g., Google scholar profile)

<https://orcid.org/0000-0003-0176-8145>

<https://scholar.google.com/citations?user=61-FEAwAAAAJ>

Examples of Previous Projects Supervised:

- Title: TH-SOF-1778-Reconstruction of 3D Face Using Dense Facial Model from a Single Image
- TH-SOF-1894-Personality Identification Through Facial Features by Using Neural Networks for Pakistani People
- Pr-sof-1573- An IOT based tracking system for agricultural vehicles.
- TH-SOF-1758-Development of Law Enforcement System for Detection of Driver's Seatbelt and Speed Estimation through Machine Vision
- PR-SOF-1961-Image Processing based approach for Testing Sphericity Compliance of Footballs
- TH-SOF-1865-Design, Implement and Benchmark a Low-Cost Optical Device for Measuring Brix and Dry Matter in Fruits
- TH-SOF-1820-Evaluation of high-resolution airborne imagery for mapping Invasive weeds in crops
- TH-SOF-1768-A Colour Based Mango Maturity Device